

In[1394]:=

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(* ::Package::*) (*=====*)
(*MAXWELL STRESS TENSOR ANALYSIS-COMPLETE*)
(*=====*) ClearAll["Global`*"];

(*-----Physical Constants-----*)
mu0 = 4 Pi * 10^-7;
m0 = 406.0;
freq = 50.0;
omega = 2 Pi freq;
turns = 100;
coilRadius = 0.02;
coilArea = Pi coilRadius^2;
z0 = 0.10;
Rcoil = 0.5;
Rload = 10.0;
Rtotal = Rcoil + Rload;
Lcoil = 3.0 * 10^-3;
tForward = 0.95;

Print["====="];
Print["MAXWELL STRESS TENSOR ANALYSIS"];
Print["====="];

(*-----Dipole Field (Full 3D)-----*)
mx[t_] := m0 * Sin[omega t];
mz[t_] := m0 * Cos[omega t];

BxDipole[t_, x_, y_, z_] := (mu0 / (4 Pi (x^2 + y^2 + z^2)^(5/2))) *
  (3 * (m0 * Sin[omega t] * x + 0 * y + m0 * Cos[omega t] * z) * x -
    m0 * Sin[omega t] * (x^2 + y^2 + z^2));

ByDipole[t_, x_, y_, z_] := (mu0 / (4 Pi (x^2 + y^2 + z^2)^(5/2))) *
  (3 * (m0 * Sin[omega t] * x + 0 * y + m0 * Cos[omega t] * z) * y - 0 * (x^2 + y^2 + z^2));

BzDipole[t_, x_, y_, z_] := (mu0 / (4 Pi (x^2 + y^2 + z^2)^(5/2))) *
  (3 * (m0 * Sin[omega t] * x + 0 * y + m0 * Cos[omega t] * z) * z -
    m0 * Cos[omega t] * (x^2 + y^2 + z^2));

(*-----Coil Field (Biot-Savart)-----*)
(*Need current from RL circuit*)
emfSource[t_] := turns * coilArea * tForward * (mu0 / (4 Pi * z0^3)) * 2 * m0 * omega * Sin[omega t];
sol = NDSolve[{Lcoil current'[t] + Rtotal current[t] == emfSource[t], current[0] == 0},
  current, {t, 0, 0.20}, MaxStepFraction -> 1 / 1000];
Icoil = current /. sol[[1]];

BcoilOnAxis[t_, z_] :=
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(mu0*Icoil[t] * turns * coilRadius^2) / (2 * (coilRadius^2 + z^2)^(3/2));

(*-----Total Field-----*)
BxTotal[t_, x_, y_, z_] := BxDipole[t, x, y, z];
ByTotal[t_, x_, y_, z_] := ByDipole[t, x, y, z];
BzTotal[t_, x_, y_, z_] := BzDipole[t, x, y, z] + BcoilOnAxis[t, z];

(*-----Maxwell Stress Tensor-----*)
MaxwellStress[t_, x_, y_, z_] := Module[{Bx, By, Bz, B2, T}, Bx = BxTotal[t, x, y, z];
  By = ByTotal[t, x, y, z];
  Bz = BzTotal[t, x, y, z];
  B2 = Bx^2 + By^2 + Bz^2;
  T = (1 / mu0) * {{Bx^2 - B2 / 2, Bx * By, Bx * Bz},
    {By * Bx, By^2 - B2 / 2, By * Bz}, {Bz * Bx, Bz * By, Bz^2 - B2 / 2}};
  Return[T];];

(*-----Compute Stress Tensor on Grid-----*)
Print["\nComputing Maxwell Stress Tensor..."];

xGrid = Range[-0.05, 0.05, 0.005];
zGrid = Range[0.02, 0.18, 0.005];
tSample = 0.005; (*Peak field time*)

stressTzz = Table[MaxwellStress[tSample, x, 0, z][[3, 3]], {x, xGrid}, {z, zGrid}];
stressTzx = Table[MaxwellStress[tSample, x, 0, z][[3, 1]], {x, xGrid}, {z, zGrid}];
stressTzy = Table[MaxwellStress[tSample, x, 0, z][[3, 2]], {x, xGrid}, {z, zGrid}];

Print["Grid size: ", Length[xGrid], " x ",
  Length[zGrid], " = ", Length[xGrid] * Length[zGrid], " points"];

(*-----Calculate Net Force on Coil-----*)
forceZ = 0;
For[i = 2, i < Length[xGrid], i++,
  For[j = 2, j < Length[zGrid], j++, dx = xGrid[[i + 1]] - xGrid[[i - 1]];
    dz = zGrid[[j + 1]] - zGrid[[j - 1]];
    forceZ = forceZ + stressTzz[[i, j]] * dx * dz;]];

Print["\n===== MAXWELL STRESS TENSOR RESULTS ====="];
Print["Peak T_zz: ", Max[Abs[stressTzz]], " N/m^2"];
Print["Peak T_zx: ", Max[Abs[stressTzx]], " N/m^2"];
Print["Peak T_zy: ", Max[Abs[stressTzy]], " N/m^2"];
Print["Net force on coil (z-direction): ", forceZ, " N"];

(*-----Stress Distribution Statistics-----*)
Print["\nStress Distribution Statistics:"];
Print[" Mean T_zz: ", Mean[Flatten[stressTzz]], " N/m^2"];
Print[" Std T_zz: ", StandardDeviation[Flatten[stressTzz]], " N/m^2"];

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Print["  Max T_zz: ", Max[Flatten[stressTzz]], " N/m^2"];
Print["  Min T_zz: ", Min[Flatten[stressTzz]], " N/m^2"];

(*-----Visualization-----*)
p1 = ListPlot3D[stressTzz, AxesLabel → {"x", "z", "T_zz"},
  PlotLabel → "T_zz Component", ColorFunction → "Rainbow", ImageSize → 400];

p2 = ListPlot3D[stressTzx, AxesLabel → {"x", "z", "T_zx"},
  PlotLabel → "T_zx Component", ColorFunction → "Rainbow", ImageSize → 400];

Print["\nStress tensor plots generated."];
Print["====="];
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MAXWELL STRESS TENSOR ANALYSIS
=====

Computing Maxwell Stress Tensor...
Grid size: 21 x 33 = 693 points

===== MAXWELL STRESS TENSOR RESULTS =====
Peak T_zz:  $1.02478 \times 10^7$  N/m^2
Peak T_zx:  $9.93412 \times 10^6$  N/m^2
Peak T_zy: 0. N/m^2
Net force on coil (z-direction): 1064.43 N

Stress Distribution Statistics:
  Mean T_zz: 42021. N/m^2
  Std T_zz: 661615. N/m^2
  Max T_zz:  $6.71713 \times 10^6$  N/m^2
  Min T_zz:  $-1.02478 \times 10^7$  N/m^2

Stress tensor plots generated.
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